

SRIVARDHAN BABA VEMULA

Open to Relocate Anywhere in the USA — srivardhanbaba@gmail.com — +1-716-604-4929 — Portfolio — LinkedIn — GitHub — Tableau

Professional Summary

Software Engineer with **3+ years** of experience building backend applications, REST APIs, cloud services, and data-intensive software. Experienced with Python, Java, SQL, AWS, Azure, Docker, distributed systems, database design, testing, and CI/CD. Contributing to AI software and deepfake detection research under **Prof. Siwei Lyu** (IEEE Fellow, Director — UB Institute for AI & Data Science).

Technical Skills

Languages	Python, Java, C++, SQL, R, Bash, Unix Shell Scripting
Backend & APIs	Flask, FastAPI, REST APIs, API Integration, Microservices, JSON, HTTP
Software Engineering	Object-Oriented Programming, Data Structures & Algorithms, Unit Testing, Integration Testing, Pytest, Git, GitHub, Docker, Linux, CI/CD
Cloud & Systems	AWS (S3, EC2, Lambda, CloudWatch), Azure (ADF, Azure SQL), Apache Kafka, PySpark, Hadoop, Distributed Systems
Databases & Data	PostgreSQL, MySQL, Microsoft SQL Server, Snowflake, Google BigQuery, Database Design, Indexing, Query Optimization, ETL/ELT, Data Modeling
ML & AI	Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, XGBoost, Feature Engineering, Model Evaluation, MLflow, NLP, Generative AI
Quality & Tools	Great Expectations, Airflow, Monitoring, Logging, Tableau, Power BI, Excel Power Query

Education

University at Buffalo — M.S. Data Science Aug 2025 – Dec 2026

Coursework: Machine Learning, Statistical Learning, Data Mining, Predictive Analytics, Probability Theory, Database Systems, Numerical Computing, Applications of Data Science: Industry Overview

Jawaharlal Nehru Technological University Hyderabad (JNTUH) — B.Tech. Computer Science Aug 2019 – Jul 2023

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Organization & Architecture, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Machine Learning, Cloud Computing, Distributed Systems, Advanced Algorithms, Computer Security, Design & Analysis of Algorithms

Experience

Graduate Research Assistant — Software Engineering & AI Systems Jan 2026 – Present

University at Buffalo | UB Media Forensic Lab | Advisor: Prof. Siwei Lyu, IEEE Fellow, AI & DS Dept. Director

- Built **Python data-processing components** for **50K+ multimodal records** stored in AWS S3, cutting preprocessing errors by **45%** for the Deep-O-Meter research platform.
- Optimized **PySpark workloads** across **100GB+ datasets** through partitioning and broadcast joins, shortening execution time by **60%**.
- Engineered **Flask REST APIs** for deep learning model inference, lowering response latency from **580ms to 378ms** while increasing throughput by **28%**.
- Evaluated **5+ deep learning architectures** through standardized training and validation workflows, raising synthetic-media classification F1-score by **12%**.
- Streamlined experiment tracking and model comparisons across **10+ configurations** with MLflow, cutting validation effort by **30%**.

Senior Software Engineer

Sep 2024 – Jul 2025

Broadridge Financial Solutions, India

- Automated **Python workflows** connecting AWS S3, SFTP, and enterprise systems; processed **200+ files daily**, increasing throughput by **40%** and eliminating **15 hrs/week** of manual processing.
- Integrated **Sterling File Gateway** with **5+ enterprise systems**, introducing validation and exception routing that lowered cross-system errors by **35%**.
- Established monitoring and recovery mechanisms with **CloudWatch, PagerDuty, and Airflow** across **12+ production workflows**, maintaining **99.8% uptime** and shortening incident resolution time by **25%**.
- Refactored **20+ SQL queries** using CTEs and window functions and redesigned **4 database schemas**, decreasing query latency by **30%** and storage overhead by **20%**.
- Instituted automated tests and deployment checks across **15+ workflows**, cutting post-release defects by **40%**.

Software Engineer

Aug 2022 – Aug 2024

Freelance, Remote

- Created **Python and SQL applications** processing **500K+ records** across **10+ data sources**, accelerating execution by **45%**.
- Restructured relational schemas and indexing strategies for **8 datasets**, achieving **60% faster query performance** across 3 enterprise clients.
- Delivered **15+ client applications and reporting interfaces** using Python and SQL, translating requirements into reusable software components.
- Strengthened production workflows with **application-level validation and error recovery**, maintaining **99.5% accuracy** across **2M+ records**.

Projects

- **Customer Analytics & Churn Prediction** (*Scikit-Learn, XGBoost*): Trained classifiers achieving **85% accuracy** via feature engineering and cross-validation.
- **Predictive Analytics Framework** (*Python, XGBoost, SQL*): Developed regression and classification models on multi-source datasets; improved prediction accuracy by **18%** using feature engineering, hyperparameter tuning, and cross-validation.
- **Deepfake Detection Research** (*PyTorch, TensorFlow, Computer Vision*): Developed deep learning classification models for synthetic media detection using feature extraction and model evaluation techniques; evaluated 5+ architectures and improved classification F1-score by **12%**.
- **Azure Cloud Data Warehouse** (*ADF, Snowflake, SQL*): Unified **6 sources** into Snowflake transforming **50M+ records/month**; shortened load SLA from **4 hrs to 45 min**.

Certifications

- **Cloud Platforms:** AWS Essentials for Developers · Azure Fundamentals (AZ-900) · Azure AI Fundamentals (AI-900)
- **Data Systems & Analytics:** IBM Data Science Professional · IBM Data Analyst Professional · IBM Data Visualization Core
- **Generative AI Systems:** Microsoft & LinkedIn Career Essentials in Generative AI Foundation